

Machine Learning–Driven Materials Design for Additive Manufacturing

Sponsored By:
Arezoo Zare, Ph.D

Faculty Mentor:
Dr. Parteek Kumar, Ph.D

By:
**Anh Duc Duong, Ryan Calimlim,
Albert Li**

School of Electrical Engineering & Computer Science
Washington State University
Pullman, Washington
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Team Introduction

Dear Arezoo Zare,

Our team is excited to meet with you and discuss the details of the " Machine Learning–Driven Materials Design for Additive Manufacturing " project. This document will give you a brief introduction to each of our team members. The following pages outline each team member's training, skills, and relevant experience. We want to thank you for this opportunity. We are excited to work and learn with you on this project.

Best Regards,

Anh Duc Duong, Ryan Calimlim, Albert Li

Albert Li
(425) 829-3177
albertli2019@outlook.com
github.com/albertli19
10020 128th Ave NE Kirkland, WA 98033

Education

Case Western Reserve University 2020-2022
Cleveland, OH
Washington State University 2024-Present
Pullman, WA
B.S. in Computer Science

Projects

Safer Ride – Case Western Reserve University, 2021

- Developed and upgraded software to manage on campus safe rides using Java
- Programmed algorithm to reduce driver and passenger wait times

Min Max Alpha Beta Pruning Optimization Algorithm – Case Western Reserve University, 2021

- In Introduction to AI, coded algorithm with heuristics to optimally move pieces on board
- Implemented min max alpha beta pruning as well as an A* heuristic and STRIPS to program enemy movement as well as player movement

IO Game – Lakeside High School, 2018

- Developed a fully working online multiplayer top-down shooter io game using JavaScript
- Used NodeJS for multiplayer

Online Marketplace – Washington State University, 2025

- Developed an interface and database for managing an online marketplace similar to Amazon, using SQL with Java to build a usable website for both vendors and customers.

Experience

Soccer Referee – East King County Soccer Referees Association, 2015-2021

- Worked as both a side referee and center referee
- Had to control all players and manage disruptive parents
- Required to take lead and single-handedly manage games

E-Filing Specialist – ABC Legal, March 2023 - August 2023

- Formatted, edited, and filed documents electronically
- Worked with PDFs, Word, and Excel documents

Technical Skills

Java * Python * HTML * CSS * SQL * C * Excel

RYAN CALIMLIM
(310) 846-7114 | realimlim113@gmail.com

SUMMARY OF QUALIFICATIONS

- Strong understanding of C/C++/C# and Engineering concepts through coursework and projects
- 10+ years of experience communicating effectively in a team (Boy Scouts, sports teams, class, work, etc.)
- Intense interest in the application of Computer Science and Software Engineering in the aerospace field

EDUCATION

Washington State University, Bachelor of Science in Computer Science Expected Dec 2026

- Current GPA: 3.38
- Relevant Coursework: Engineering Administration, Project Management, Data Mining, Cybersecurity and Cryptography, Advanced Data Structures (C/C++), Systems Programming (C/C++), Computer Architecture, Software Engineering, Database Management

SKILLS

C/C++/C#, SQL, Kernel/OS Systems Programming, NUNIT testing/debugging, Winforms, Selenium, LaTeX, Docker, Blazor, HTML, Software & Data Structure Design, TIG Welding, Soldering, CAD, Linux, Project Management, De-Escalation/Conflict Resolution, Power Tools

RELEVANT PROJECTS

Virtual Counsellor Group Project, Washington State University Spring 2025

- Designed and implemented a virtual student counsellor website for Washington State University
- Backend implemented using C#, hosted through a Docker container
- Devised a Selenium web scraper that stores the 200+ fields of study and 2000+ classes offered at WSU

Spreadsheet Program Personal Project, Washington State University

Spring 2025

- Designed and implemented a spreadsheet program like Excel using WinForms and C# (1000+ cells)
- Functionality: Sheet can save and load. Cells can add, subtract, multiply, divide, use exponents, and use parenthesis, both internally and on other cells (cell A1=1+cell A2)

Eagle Scout Community Service Project, Scouts BSA: Troop 915, California

Spring 2021

- Designed and constructed an outdoor gardening classroom including 4 planters and 2 potting tables
- Planned, budgeted, and led 20-30 people for the entire project, managing and delegating tasks
- Communicated extensively with the project recipient on project specifications and timeline

Anh Duc Duong
19aduong4@gmail.com | github.com/AnhDucDuong | 714-343-1495

Summary:

Aspiring Computer Science professional with experience in coding, web development, and collaborative problem-solving. Skilled in front-end (HTML, CSS, JavaScript) and back-end development, with a focus on implementing scalable solutions and improving performance. Adept at working in fast-paced environments and applying best practices to deliver high-quality results.

Skills

Programming languages: C/C++, Python, Java, SQL, TypeScript/JavaScript, HTML/CSS

Tools and technologies: Git, GitHub, GitLab, Linux, Visual Studio Code, Next.js, React.js, Microsoft 360

Database: MongoDB, PostgreSQL, SQLite.

Soft skills: Collaboration, Communication Skills, Problem-solving, Teamwork

Communication: Fluent in English and Vietnamese

Project:

Back-End Developer & Database Manager – Qwest

Collaborating with a team to develop **Qwest**, a social event discovery and engagement platform designed for students. The app features **real-time event feeds**, **personal profiles**, and **direct messaging**, encouraging students to connect and participate in nearby events. Users can **earn social points** for attending activities, view live updates, and interact through a dynamic community interface. Responsible for **database design**, **back-end API integration**, and **Cloud SQL management** to ensure scalability and smooth data flow across features.

Research Assistant – Hardware-Accelerated AI (Ongoing)

Collaborating with a faculty-led research project under the supervision of Professor Abhishek Moitra, focusing on **improving AI algorithms for hardware-accelerated systems**. The project explores algorithm-level optimizations to **enhance computational efficiency** and **performance on accelerator-based platforms**. Responsibilities include analysing existing AI algorithms, identifying performance bottlenecks, and contributing to the development and evaluation of more efficient algorithmic approaches. Research is currently in progress.

Education:

Washington State University, Pullman, WA

- Bachelor of Computer Science - Expected to graduate in December 2026
- GPA 3.84
- Relevant Course: Advanced Data Structures, Automata and Formal Languages, System Programming,

Everett Community College, Everett, WA

- Associate of Arts and Science (*Honors Graduated*)